1.2 Final Report

Thoa Le (a1847933)

Abstract

Introduction

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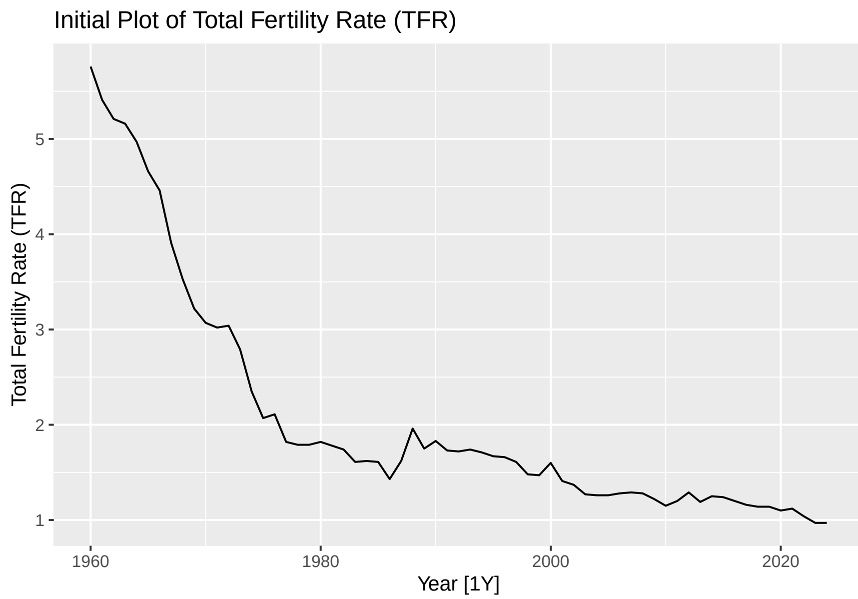
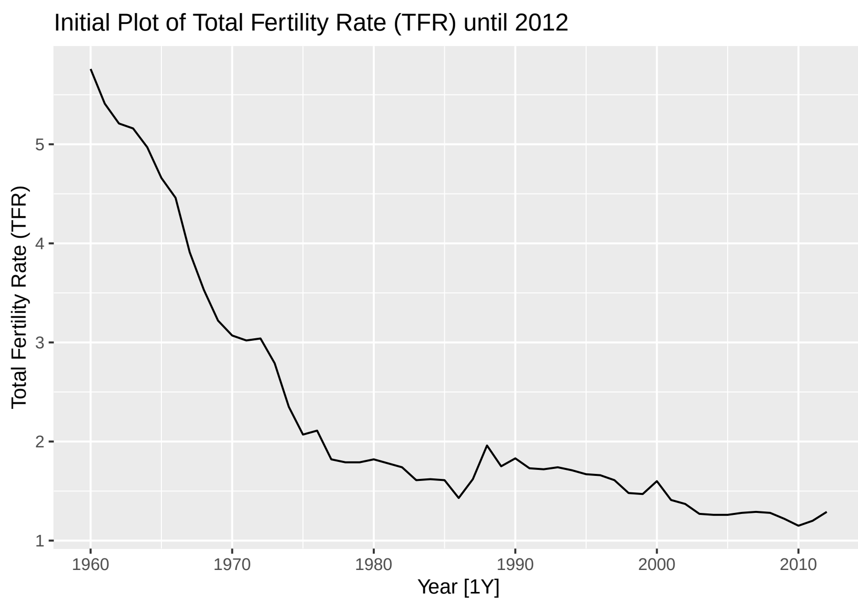
Appendices

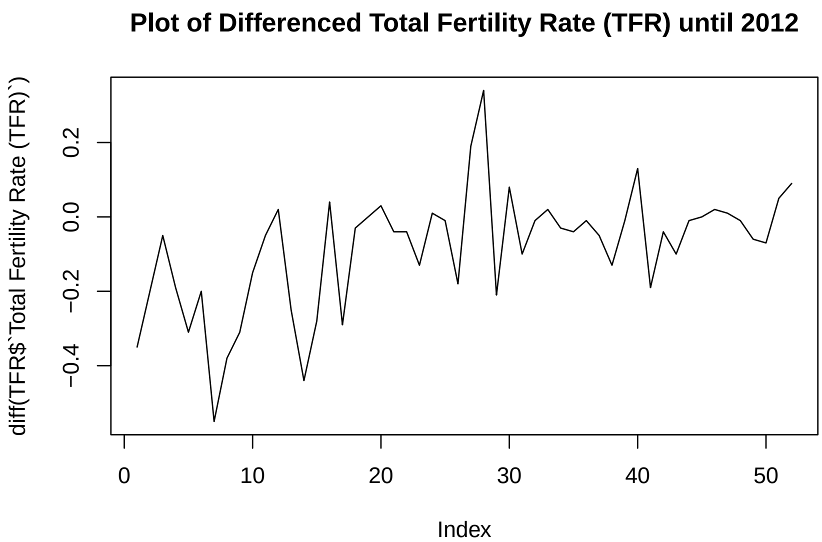
Data from 1960 to 2024 on Total Live Births (TLB) and Total Fertility Rates (TFR) was pulled from Singapore Department of Statistics (data.gov.sg). The data was then transformed into a tsibble object with three columns: Year, Total Fertility Rates (TFR) and Total Live Births.

From the exploratory data analysis, the data was visualised to identify patterns and trends over time. Since the data is yearly, classical decomposition and the X-11 method are not applicable and seasonal patterns cannot be detected. Additionally, the socio-economic and political factors influencing fertility are not included. Once the patterns are identified, appropriate models can be selected, fitted, and evaluated for forecasting.



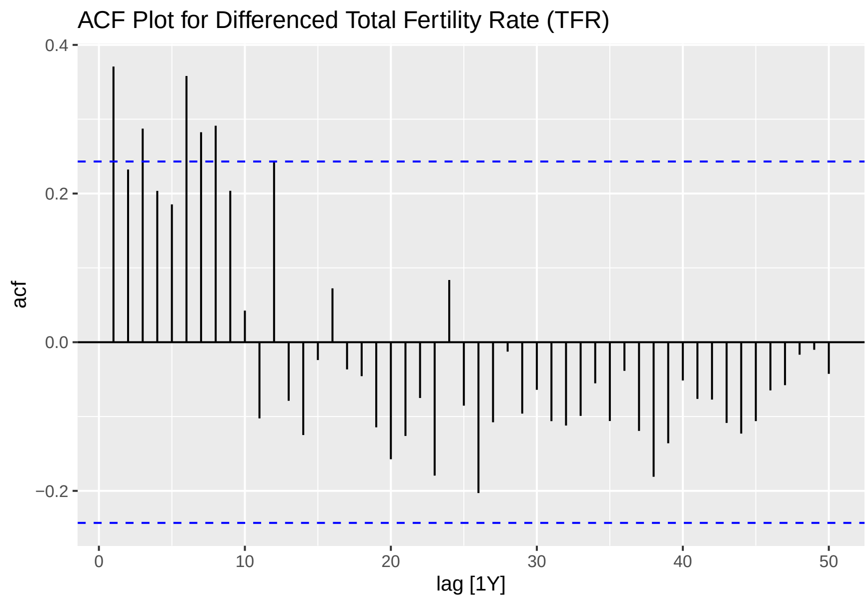


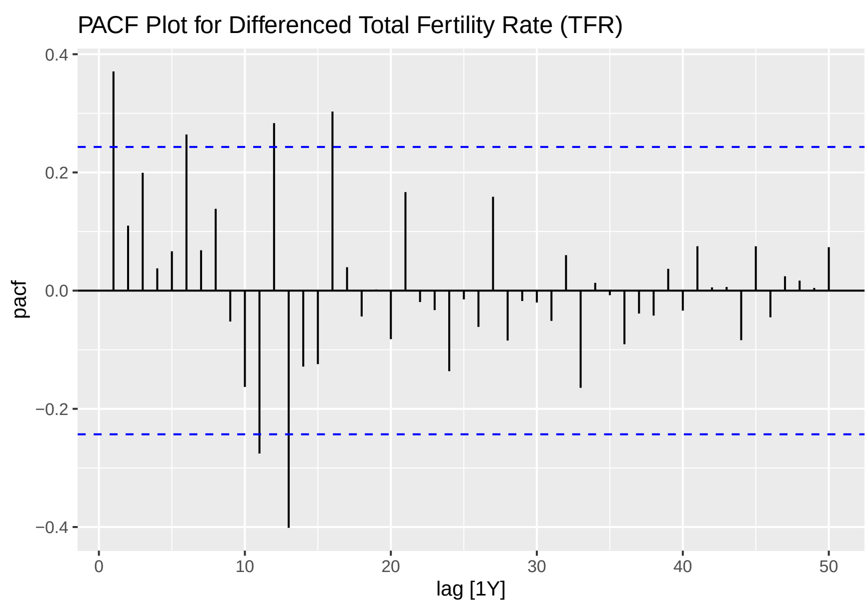
From the initial plots of TFR and TLB, both series show a long-term decreasing trend over time, with TFR exhibiting a more pronounced decline. A notable change is observed around 1980 to 1990, where both series display a temporary broad peak before continuing their downward trend. As the data is yearly, seasonality cannot be determined. While both series may suggest cyclical behaviour, the trend would need to be removed to confirm this.



By using differencing, the plots look like the trend has been removed.

KPSS test



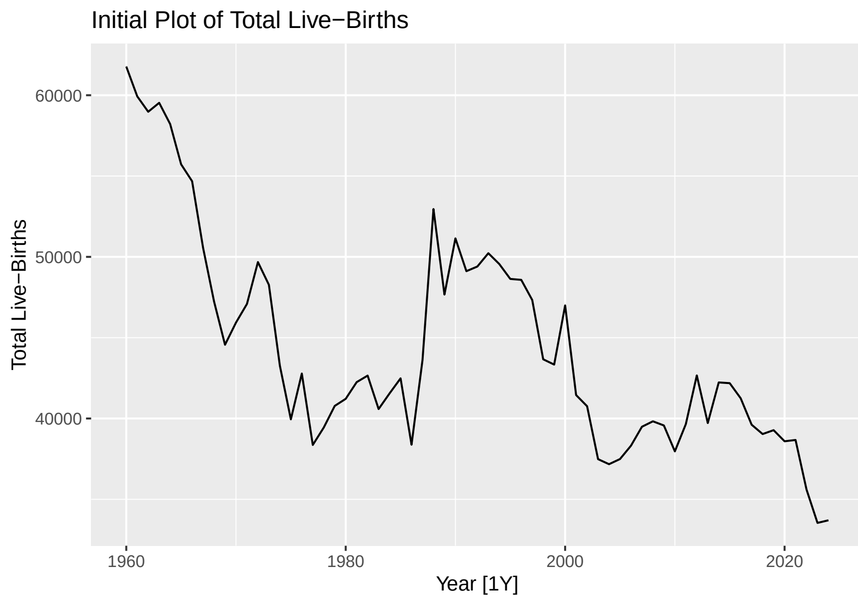


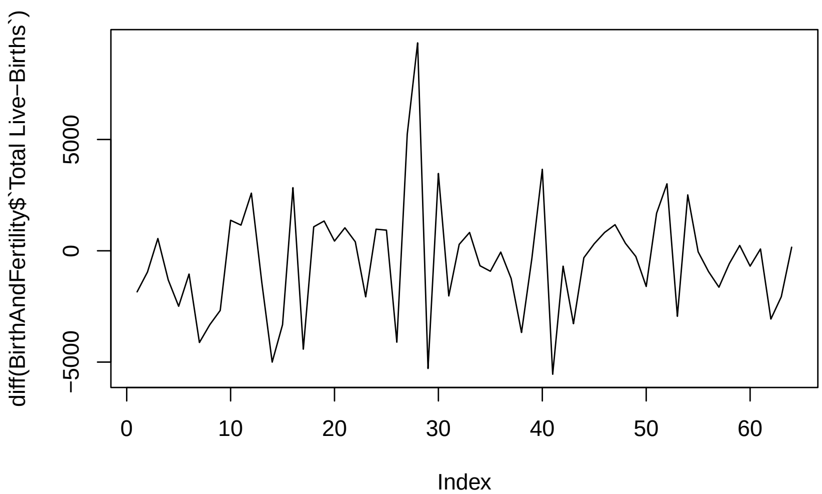
The ACF for TFR shows significant positive autocorrelations from lags 1 to 9, while TLB shows significant positive autocorrelations from lags 1 to 6, with a slow decrease in the ACF as the lag increases. This suggests that neither series is white noise, as there are multiple spikes outside the significance bounds.

The presence of a strong trend and slow decrease confirms that both TFR and TLB are non-stationary series. The absence of a sharp cut-off suggests that differencing or an autoregressive (AR) model may be suitable for this series. An extended ACF plot for TFR confirms that significant positive autocorrelations persist up to lag 10.

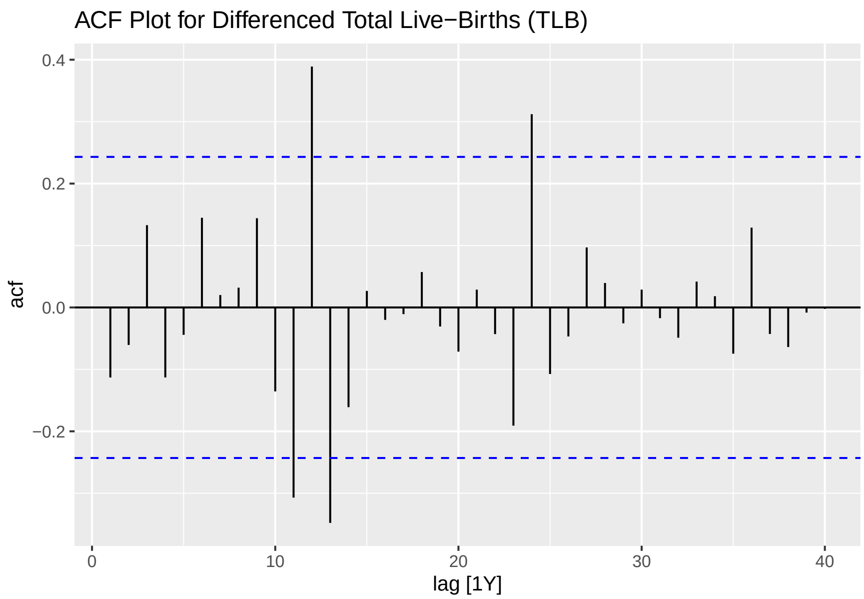
ARIMA (16,1,0)

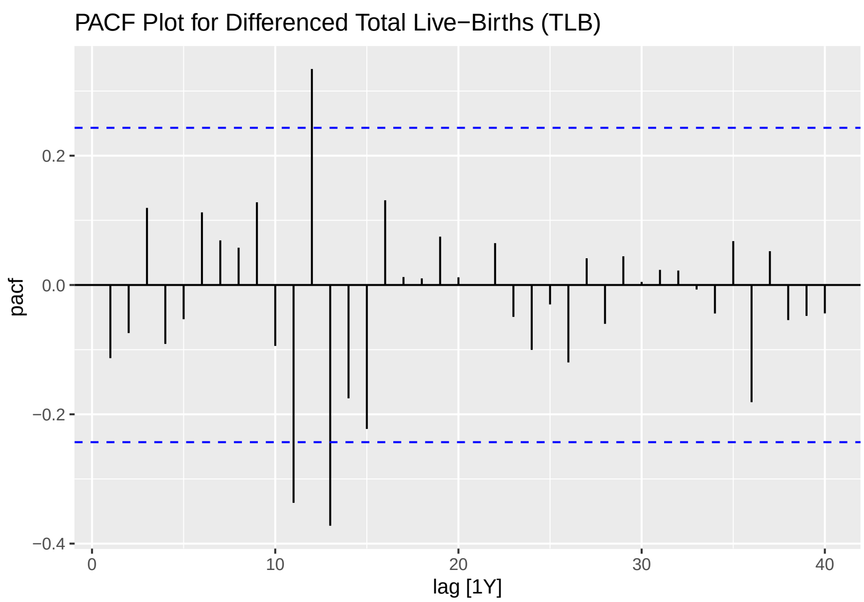
```
## Series: TFR$`Total Fertility Rate (TFR)`
## ARIMA(16,1,0)
##
## Coefficients:
##          ar1      ar2     ar3      ar4     ar5     ar6     ar7     ar8      ar9
##       0.3647  -0.1061  0.4577  -0.2586  0.2984  0.2452  0.2006  0.1578  -0.0293
## s.e.  0.1240   0.1289  0.1246   0.1128  0.0935  0.0832  0.0952  0.0928   0.0996
##          ar10     ar11    ar12     ar13     ar14     ar15    ar16
##       -0.0306  -0.3594  0.3889  -0.6048  -0.0419  -0.3361  0.5135
## s.e.   0.0973   0.0913  0.1039   0.1066   0.1350   0.1295  0.1397
##
## sigma^2 = 0.009716:  log likelihood = 48.99
## AIC=-63.99   AICc=-45.99   BIC=-30.82
```


Initial Plot of Total Live−Births
60000
50000
40000
Total Live−Births
1960
1980
2000
2020
Year [1Y]


5000
0
−5000
diff(BirthAndFertility$`Total Live−Births`)
0
10
20
30
40
50
60
Index

```
## NULL
```


ACF Plot for Differenced Total Live−Births (TLB)
acf
0.4
0.2
0.0
−0.2
0
10
20
30
40
lag [1Y]


PACF Plot for Differenced Total Live−Births (TLB)
pacf
0.2
0.0
−0.2
−0.4
0
10
20
30
40
lag [1Y]

ARIMA (13,1,0)

```
## Series: Total Live-Births
## Model: NULL model
## NULL model
```

Research Questions

Can the ARIMA model accurately forecast the Singapore's Total Live Births (TLB) and Total Fertility Rate (TFR) from 2013 to 2024 given the data from 1960 to 2012, capturing the short-term fluctuations?

Statistical Appendix

The following statistical methods were applied throughout the analysis in the order presented below.

Autoregressive Model (AR)

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

where:

y_t is the value at time t

ϕ_i are model parameters

ε_t is white noise

Models the current values using its previous values (lags)

Residuals

$$e_t = y_t - \hat{y}_t$$

y_t is the actual observed value at time t

$\hat{y}_t$ is the predicted value at time t

The error which is the difference between the actual observed value and predicted value at time t, used to check the chosen model accounts for all the patterns

Autocorrelation Function (ACF)

$$\rho_k = \frac{\mathrm{Cov}(y_t, y_{t-k})}{\mathrm{Var}(y_t)}$$

where:

ρ_k is the autocorrelation at lag k

Measures the correlation between a series y_t and lagged values y_{t-k}, which checks for trend, seasonality and whether residuals are white noise

Box-Cox Transformation

$$y^{(\lambda)} = \begin{cases} \frac{y^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \log(y), & \lambda = 0 \end{cases}$$

λ was estimated using the Guerrero method

STL Decomposition

$$y_t = T_t + R_t$$

where:

T_t is the trend component,

R_t is the remainder

Decomposes time series into trend and remainder components, usually with the seasonality component as well but the data in this analysis is yearly

Moving Average (MA)

$$MA_k = \frac{1}{k} \sum_{i=-(k-1)/2}^{(k-1)/2} y_{t+i}$$

k is the window size

In this analysis when k = 5, the moving average at time t is calculated using the 2 observations before and 2 observations after t. This process smooths out short-term fluctuations in the series

https://github.com/twalepear/TSA.git